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3004ENG — Project Management Principles

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Eris TestFlight2 Launch Campaign

Project Execution Plan

Assessment 3: Group Project Execution Plan — Group 1

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Acting as a project management consultancy engaged to deliver launch-site operations at Bowen Orbital Spaceport.

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1 Executive summary

This Project Execution Plan asks the Gilmour Executive Board to approve or decline Eris TestFlight2 *launch-site operations* at Bowen Orbital Spaceport from this section alone. TestFlight 1 cleared the tower and ended at T+14 s after thrust loss in two first-stage hybrid motors, from electrical and thermal pump faults (Smith, 2025; Gilmour Space Technologies, 2025; Dougherty, 2026). Group 1 is a project-management consultancy. In-scope work is pad operations: clean-room receipt, 23 m stacking, cryogenic GSE, a cold Wet Dress Rehearsal, a five-second static fire of the four first-stage hybrid engines, range clearance, lift-off, telemetry handover and closeout. Factory manufacture, engine redesign and freight to the receiving gate stay with Gilmour and sit outside this baseline.

The work estimate is \$1,500,000. Contingency is \$81,050 (5.40% of base): \$37,920 of residual expected monetary value on eleven cost-bearing risks plus a \$43,130 itemised in-scope residual-uncertainty allowance. Authorised *BAC* is \$1,581,050. Contingency may be consumed; nothing beyond *BAC* is authorised. Delivery is a managing-contractor / design-and-construct hybrid: Gilmour keeps propulsion intellectual property, vehicle integration and the launch licence; civil and cryogenic GSE packages are let design-and-construct with an interface-control freeze and liquidated damages.

The campaign uses a seven-day pad calendar from 10 August to 1 December 2026 because LOX load, static fire and *T-0* do not pause for weekends. Permitting and Juru mobilisation are already on the clock from 10 August. Gate 1 on 2 October locks this PEP and remaining pad spend; it does not start a campaign that has not begun. Stacking completes on 15 October 2026 (M-4 / A-123). Cold Wet Dress Rehearsal completes on 24 October (A-132). The five-second static fire completes on 30 October 2026 (M-5 / A-133). Launch Readiness Review is on 10 November 2026 (M-6 / A-141). Baseline *T-0* is **15 November 2026** (M-7 / A-143), inside the charter's Q1 2027 outer window. Telemetry handover is on 20 November 2026 (M-8 / A-144), with at least 98.5% of packets. Closeout is on 1 December 2026 (M-9 / A-145). A five-day pull-in to 10 November exists only if R-14 is realised, at a contingent \$21,000 draw on the reserve. It is not the baseline.

Three register items can stop *T-0* or close the heritage case: R-01, R-12 and R-14. R-01 is a residual pump-insulation defect: static-fire auto-abort on A-133 and a 12–14 day critical-path recycle. R-12 is the TestFlight 1 in-flight mode: thrust loss in two first-stage hybrid motors after *T-0*, as at T+14 s, ending the heritage objective. R-14 is an ASA flight-authorisation or CASA NOTAM refusing the 15 November slot. Occupancy is held at the pad occupancy cap of twelve after an unmitigated peak of fourteen. The organic crew is eight.

It is recommended that the Executive Board approve this Project Execution Plan, baseline the campaign at \$1,581,050 AUD (inclusive of \$81,050 contingency), lock remaining pad spend at Gate 1 on 2 October 2026, and hold T-0 on 15 November 2026 inside the charter's Q1 2027 outer window.

2 Refined project overview

2.1 Background and purpose

This is the Week 11 reading of a campaign chartered on 7 August 2026. Gilmour Space Technologies flew Eris TestFlight 1 from Bowen in July 2025. The vehicle cleared the tower and was terminated at T+14 s after thrust loss in two first-stage hybrid motors (Smith, 2025; Gilmour Space Technologies, 2025; Australian Space Agency, 2025). TestFlight 2 is the late-2026 follow-on, inside the charter's Q1 2027 outer window (Dougherty, 2026). Group 1 plans and controls *launch-site operations*: cleanroom receipt, 23 m stacking, cryogenic GSE, a cold Wet Dress Rehearsal, a five-second static fire of the four first-stage hybrid engines, range clearance, lift-off on 15 November 2026, handover of at least 98.5% of telemetry within five days, and closeout. Manufacturing, engine research, freight to the gate, and pump redesign remain Gilmour factory work (Project Management Institute, 2021; Kerzner, 2017). Orbital insertion is Gilmour's vehicle-success test, not a consultancy finish-line metric.

The Bowen facility licence has been held since March 2024, so ASA flight authorisation, CASA airspace and GBRMPA permissions gate hot fire, not stacking (Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019; Australian Space Agency, 2025). The Australian Space Agency stays Keep Satisfied, with filings 60 days prior, as the charter scored it; that filing sits on A-113 with the CASA NOTAM. The Juru people stay High power / High interest, Manage Closely. That class is unchanged. The new control is monitors with stop-work authority on the network before Wet Dress Rehearsal. The payload sponsor stays Keep Informed and witnesses LRR on A-141.

2.2 Scope

Scope is the 100% rule on those site operations (Project Management Institute, 2021). Included: campaign governance and EVM (WBS 1.1), pad logistics and GSE (1.2), pad testing (1.3), and flight, launch and closeout (1.4). Seventeen Level-3 packages in Figure 2.1 are the cost accounts. Nineteen schedule bars—those seventeen, the 1.2.1 split into A-121 and A-122, and external A-100—are the network in Table 3.1. Excluded: factory manufacture, engine R&D, line-haul to the gate, and any unidentified Level-3 box that would mix factory product into a site baseline. A-100 is a zero-dollar external driver: it is launch-critical because freight gates stacking, but this consultancy does not control it. It is omitted from Table 4.1. Design defects found on the pad are packaged back to R&D. Only a pre-kitted spare may be swapped on A-134.

2.3 Charter baseline versus PEP refinement

A refined overview is not a restated charter (Kerzner, 2017). Table 2.1 lists every material change, including two Assessment 1 date collisions: M-4 locked to A-123 on 15 October 2026, and M-8

locked to A-144 on 20 November. It also records the remapping of Objectives 4 and 5, LRR attendees, and the Week 8 charter review (no roles or DoA changed).

Table 2.1: Charter baseline versus PEP refinement. A1 wording is the Week 4 charter. A3 dates and dollars are the locked YAML values used in Tables 3.1, 4.1 and 6.1.

Project element	A1 charter (Week 4)	A3 PEP (Week 11)	Why it changed
M-4 stacking date	Objective 1: pad integration 15 Oct. Milestone M-4: stacking 28 Oct.	M-4 locked to 15 Oct 2026 (A-123 <i>EF</i>).	Removes the internal collision. Creates a 15-day pad-test buffer before the 30 Oct hot-fire gate.
M-8 handover date	“5 days post-launch” and milestone date 22 Nov (7 days after 15 Nov).	M-8 = 20 Nov 2026 (A-144).	Aligns the milestone with Objective 5 and the 5-day rule.
Cost baseline	“Manage within the allocated operational budget.” No figure.	Bottom-up \$1,500,000 + \$81,050 contingency (residual EMV \$37,920 + RES \$43,130).	Crane hire, cryogenic crews, and LOX delivery cannot be controlled without a numbered baseline.
Permit vs stacking logic	M-3 flight permits 25 Oct were predecessors of M-4 stacking (then 28 Oct).	Facility licence already held (Mar 2024). A-113 gates A-133 static fire, not stacking. Permits <i>EF</i> 24 Oct; hot fire <i>ES</i> 25 Oct.	Stacking is a site-licence activity. Hot fire is a range activity. This also creates a second critical path.
Juru engagement	High power / high interest, Manage Closely, consultation and monitoring.	Same power/interest (High / High, Manage Closely). Embedded monitors with stop-work authority and a cultural hold point before WDR (A-114 → A-132).	A1 strategy was correct in class; the PEP makes it an enforceable pad control, not a comms plan.
Testing sequence	WDR and 5-second static fire as one Objective 3 output by 30 Oct.	A-132 WDR (cold) 16 Oct–24 Oct, then A-133 static fire 25 Oct–30 Oct. LRR waits on A-134 diagnostics.	Hot fire must wait for flight permits. LRR cannot start on incomplete post-fire data.
Risk governance	Five high-level threats, no $P \times I$ money.	18 campaign-specific risks, Cause→Event→Effect, EMV \$37,920 + RES \$43,130.	Generic rows cannot size contingency or point at a WBS line.
Resource peak	Roles listed; no demand vs capacity.	Organic crew 8 , pad HSE cap 12 , unmitigated peak 14 , mitigated peak 12 .	A1 could not show where the plan exceeds capacity.
Regulator and payload	ASA High/Low Keep Satisfied, file 60 days prior. Payload sponsor Med/High Keep Informed, witness LRR.	ASA remains Keep Satisfied. CASA issues the airspace NOTAM (R-14). Payload sponsor still witnesses LRR on A-141 (HP-4). No 10–12 Nov geometry is used.	A1 named ASA, not Townsville, as the high-power regulator. The PEP keeps that filing rule and uses CASA as the airspace instrument.

Table 2.1: Charter baseline versus PEP refinement (continued).

Project element	A1 charter (Week 4)	A3 PEP (Week 11)	Why it changed
Success criteria	Q1 2027 window; no funds beyond contingency; 98.5% telemetry in 5 days; zero LTI/GBRMPA/Juru; defects to R&D.	T-0 15 Nov inside A1's Q1 2027 window (2-day in-flight trigger); $AC \leq BAC$; 98.5% telemetry in 5 days; LRR first submission; four-engine 5 s fire; zero LTI/GBRMPA/Juru; defects to R&D. LEO is Gilmour's test, not the consultancy's.	A1 tests stay. The PEP numbers cost and tightens in-flight control inside Q1 2027. LEO is not this team's finish line.
Objectives 4 and 5	Objective 4: pass LRR by 10 Nov. Objective 5: November 2026 window and 98.5% telemetry.	Objective 4 is T-0 on 15 Nov (M-7) inside Q1 2027, with a 2-day in-flight trigger. Objective 5 is 98.5% handover on 20 Nov (M-8) and closeout on 1 Dec. LRR remains M-6 / first-submission success, not a numbered objective.	The numbered objectives now match the gates this consultancy controls. LRR is still HP-4; it is no longer counted twice as both Obj 4 and M-6.
LRR attendees	M-6: Launch Director, Range Safety Officer and Juru Land Liaisons review. Payload sponsor witnesses LRR.	HP-4 release is Launch Director, RSO, Juru Officer, Abdul and Ziyad. Payload sponsor remains a witness, not a signatory.	A1 named the review board. The PEP keeps those three on the hold and does not promote the payload sponsor from witness to signatory.
Week 8 charter review	Team will review this charter in the Week 8 workshop and revise roles if necessary.	Week 8 workshop reopened the charter. No roles, DoA or contribution split were renegotiated.	The brief asked teams to record renegotiations. There were none; omitting the row would look as if the review was skipped.

Building the network taught the non-date rows. A-113 finishes 24 October and gates static fire, not stacking: that second $TF = 0$ path is why crashing A-123 cannot move 15 November. Locking M-4 to 15 October restores the fifteen-day cold-flow window before the 30 October hot-fire gate. Juru stays Manage Closely, but A-114 now feeds A-132 with stop-work authority. The charter's five-second static fire of four hybrid engines remains; it is sequenced after cold WDR rather than fused with it. Cost, risk and occupancy are now numbered: \$1,500,000 plus \$81,050 of residual-EMV-plus-RES contingency, eighteen cause–event–effect rows, organic crew eight against a pad cap of twelve.

2.4 Objectives and success criteria

Objectives are auditable against the gates in Sections 3–7. Objective 1 is stacking complete on 15 October 2026 (M-4 / A-123). Objective 2 rewrites the Week 4 qualitative budget: *Deliver*

launch-site operations within the \$1,500,000 AUD base estimate, managing an authorised \$81,050 AUD contingency sized from residual EMV plus a RES uncertainty allowance, audited at Post-Stacking (M-4), Post-Static Fire (M-5), and Post-Launch (M-7). Objective 3 separates that 30 October gate: A-132 is cold WDR (16–24 October); A-133 is the five-second static fire of the four first-stage hybrid engines (25–30 October). Objective 4 is $T-0$ on 15 November 2026 (A-143 / M-7), the baseline slot inside the charter’s Q1 2027 outer window, with an in-flight slip trigger of more than two days. Objective 5 is handover of at least 98.5% of decrypted telemetry by 20 November (A-144 / M-8) and closeout on 1 December (A-145 / M-9).

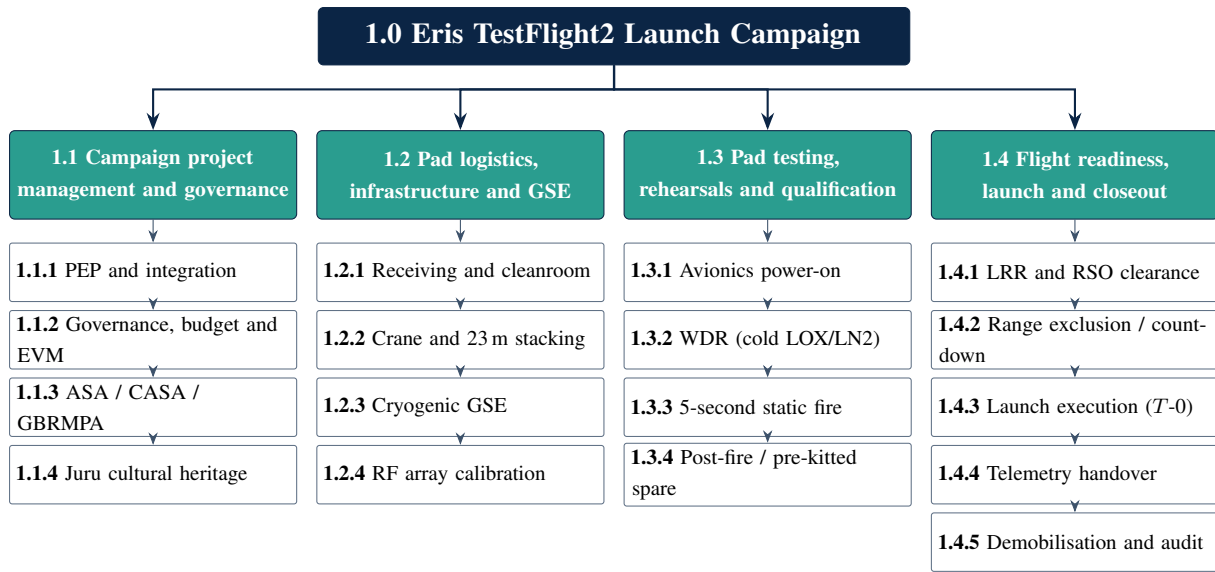
Finish-line success is the pair used in the earned-value framework (Project Management Institute, 2021; Kerzner, 2017): final $AC \leq \$1,581,050$ AUD, and $T-0$ on 15 November 2026 with that > 2 day trigger. Contingency may be consumed; nothing beyond BAC is authorised. LRR must pass on the first formal submission. HSE success stays as chartered: zero lost-time injuries, zero GBRMPA breaches, and zero Juru heritage breaches. Design defects return to R&D. LEO insertion is Gilmour’s test, not this consultancy’s.

2.5 Work breakdown structure

The work breakdown structure in Figure 2.1 is the 100% rule applied to launch-site operations at Bowen Orbital Spaceport (Project Management Institute, 2021). Four Level-2 workstreams—campaign governance, pad logistics, pad testing, and flight/closeout—decompose into seventeen Level-3 packages. Every scheduled activity except A-100 maps to exactly one of those packages, and every dollar in Table 4.1 sits in exactly one of them. A-100 is annotated beside the tree rather than inside it: the vehicle receipt window is an external zero-dollar wait that starts when out-of-scope freight reaches the gate. It belongs on the network because it drives stacking (A-122), but it is not a cost account.

Package 1.2.1 is one cost account with two schedule bars (A-121 site mobilisation and A-122 receiving inspection) because receiving cannot start until both the cleanroom and the vehicle exist. The remaining sixteen Level-3 codes are one-to-one with Table 3.1. Factory manufacture, engine research and development, and line-haul to the receiving gate are excluded.

The 7 August charter does not contain a notice to proceed for field work. Permitting (A-113) and Juru mobilisation (A-114) start on 10 August because those clocks have to run before hot fire. Physical pad occupancy before Gate 1 (A-111, 2 October) is a PEP choice, not a charter clause. What Gate 1 locks is the cost and schedule baseline.



EXT A-100 — not a WBS cost account. External \$0 freight driver into A-122; omitted from Table 4.1.

Figure 2.1: Work breakdown structure for Eris TestFlight2 launch-site operations. Seventeen Level-3 packages are the cost accounts in Table 4.1. A-100 is an external schedule bar, not a seventeenth-plus cost line.

3 Schedule management

The campaign schedule converts the WBS into a time-phased baseline for board approval. Dates are derived from a finish-to-start activity-on-node network on a seven-day pad calendar from 10 August to 1 December 2026: LOX load, static fire and $T-0$ do not pause for weekends. Approved $T-0$ remains 15 November 2026, inside the charter’s Q1 2027 outer window. A five-day pull-in to 10 November is a contingent response if R-14 is realised (ASA flight authorisation or a CASA NOTAM refuses the 15 November slot). It is not the baseline.

3.1 Network logic and float analysis

Precedence is modelled as activity-on-node with finish-to-start links and zero lag (Project Management Institute, 2021; Kerzner, 2017). Textbook discrete-time notation writes $EF = ES + D$ when EF is exclusive. On the Bowen calendar every displayed date is inclusive, so the implemented identities are

$$ES_i = \max_{j \in P_i}(EF_j) + 1, \quad EF_i = ES_i + D_i - 1, \quad (1)$$

$$LF_i = \min_{k \in S_i}(LS_k) - 1, \quad LS_i = LF_i - D_i + 1, \quad (2)$$

with P_i and S_i the predecessor and successor sets of activity i , and with ES equal to the campaign start where P_i is empty. Total float and free float then follow as

$$TF_i = LS_i - ES_i = LF_i - EF_i, \quad FF_i = \min_{k \in S_i} (ES_k) - EF_i - 1, \quad (3)$$

taking $FF_i = TF_i$ when S_i is empty. Launch-critical means $TF_i = 0$ on a path that sets $T-0$ or closeout. Table 3.1 is the full pass and matches the locked register.

The forward pass produces two $TF = 0$ chains that remain independent until hot fire. The vehicle path is A-100 (external freight driver; not consultancy-controlled), then A-122, A-123, A-132 and A-133. The permit path is A-113 into A-133. RF calibration (A-125) is a finish-to-start predecessor of A-133, so the array is live before hot fire; it is not launch-critical. After A-133 the tail A-134–A-145 is a single critical chain. A-111 finishes on 2 October by an imposed Must-Finish-On (Gate 1) and is not launch-critical. The Week 4 charter approved the document, not field mobilisation.

Path convergence is two staged merges. Wet Dress Rehearsal (A-132) cannot start until stacking (A-123), cryogenic GSE (A-124), avionics power-on (A-131) and Juru monitors (A-114) are complete—the technical merge. Static fire (A-133) then waits on A-132, flight permits (A-113) and RF calibration (A-125)—the regulatory and range-instrument merge. Launch Readiness Review (A-141) waits on post-fire diagnostics (A-134), not on A-133. Signing a flight-clearance gate on incomplete hot-fire data would break the hold-point logic required of an AS9100D system (SAE International, 2016).

Feeder float is not interchangeable. A-131 has one day into A-132; A-124 has three. A-125 has 39 days and is parked in August–September so it does not sit on the October pad peak; moving it does not change 15 November. A-121 has $TF = 3$ but $FF = 0$, so delay is absorbed before $T-0$ only by consuming A-124's float. Total float protects $T-0$; free float protects the successor. Figure 3.1 draws the links.

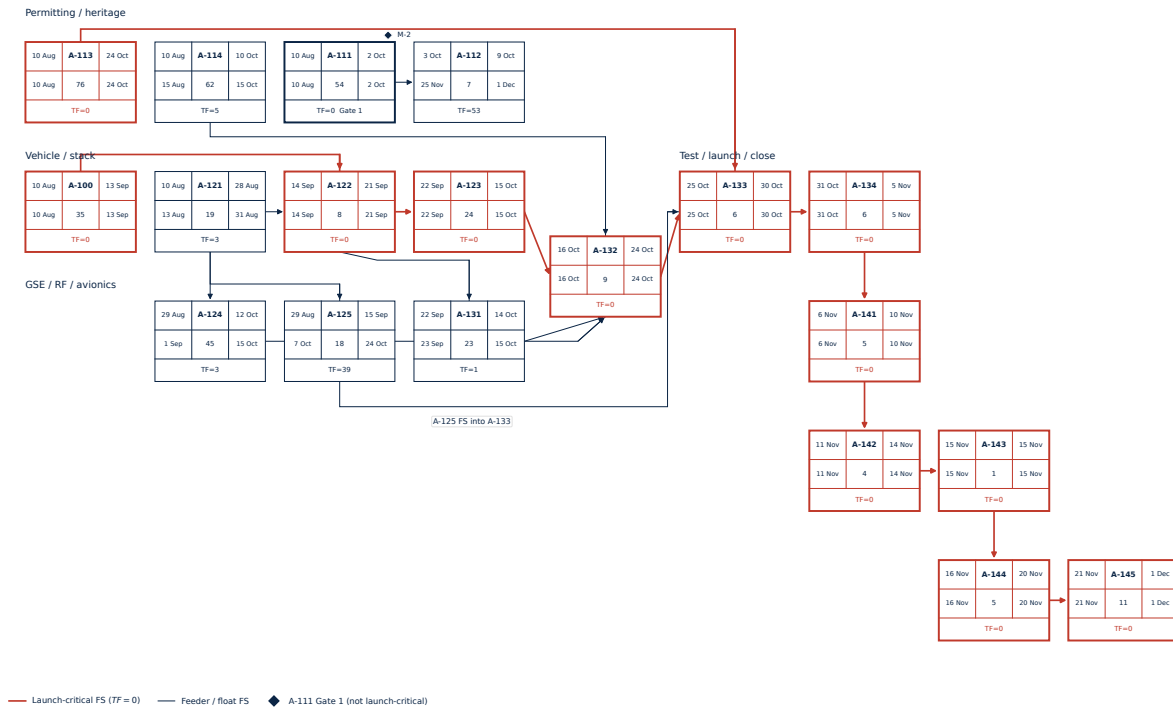


Figure 3.1: Activity-on-node network for the Bowen campaign. Node bands are $ES / ID / EF$, $LS / \text{duration} / LF$, and TF . Crimson arrows are launch-critical finish-to-start links. The vehicle path and the A-113 permit path merge at A-133; A-125 is the labelled south-routed feeder into the same merge; A-141 waits on A-134, not on A-133. A-111 is Gate-1 constrained and is not launch-critical. A-100 is an external driver.

Table 3.1: Master campaign activity precedence and float register. Dates are inclusive. $FF = \min(ES_{\text{succ}}) - EF - 1$ days, or $FF = TF$ if there is no successor. A-111 $TF = 0$ is an imposed Gate 1 finish, not launch-critical.

ID	WBS	Activity	D	Pred.	ES	EF	LS	LF	TF	FF	CP
A-100	EXT	Vehicle receipt (external driver)	35	—	10 Aug	13 Sep	10 Aug	13 Sep	0	0	Y
A-111	1.1.1	PEP and integration baseline	54	—	10 Aug	2 Oct	10 Aug	2 Oct	0*	0	Gate
A-112	1.1.2	EVM control stand-up	7	A-111	3 Oct	9 Oct	25 Nov	1 Dec	53	53	N
A-113	1.1.3	ASA / CASA / GBRMPA permits	76	—	10 Aug	24 Oct	10 Aug	24 Oct	0	0	Y
A-114	1.1.4	Juru monitor mobilisation	62	—	10 Aug	10 Oct	15 Aug	15 Oct	5	5	N
A-121	1.2.1a	Site mobilisation and cleanroom	19	—	10 Aug	28 Aug	13 Aug	31 Aug	3	0	N
A-122	1.2.1b	Receiving and unpacking	8	A-100, A-121	14 Sep	21 Sep	14 Sep	21 Sep	0	0	Y
A-123	1.2.2	Crane and vertical stacking	24	A-122	22 Sep	15 Oct	22 Sep	15 Oct	0	0	Y
A-124	1.2.3	Cryo GSE piping and checkouts	45	A-121	29 Aug	12 Oct	1 Sep	15 Oct	3	3	N
A-125	1.2.4	RF telemetry calibration	18	A-121	29 Aug	15 Sep	7 Oct	24 Oct	39	39	N
A-131	1.3.1	Avionics power-on sweeps	23	A-122	22 Sep	14 Oct	23 Sep	15 Oct	1	1	N
A-132	1.3.2	Wet Dress Rehearsal (cold)	9	A-114, A-123, A-124, A-131	16 Oct	24 Oct	16 Oct	24 Oct	0	0	Y
A-133	1.3.3	5-second static fire	6	A-132, A-113, A-125	25 Oct	30 Oct	25 Oct	30 Oct	0	0	Y
A-134	1.3.4	Post-fire / pre-kitted spare	6	A-133	31 Oct	5 Nov	31 Oct	5 Nov	0	0	Y
A-141	1.4.1	LRR and RSO clearance	5	A-134	6 Nov	10 Nov	6 Nov	10 Nov	0	0	Y
A-142	1.4.2	Range exclusion and countdown	4	A-141	11 Nov	14 Nov	11 Nov	14 Nov	0	0	Y
A-143	1.4.3	Launch execution ($T-0$)	1	A-142	15 Nov	15 Nov	15 Nov	15 Nov	0	0	Y
A-144	1.4.4	Telemetry decrypt and handover	5	A-143	16 Nov	20 Nov	16 Nov	20 Nov	0	0	Y
A-145	1.4.5	Demobilisation and closeout	11	A-144	21 Nov	1 Dec	21 Nov	1 Dec	0	0	Y

3.2 Schedule baseline, Gantt visualisation and milestone integrity

The dates in Table 3.1 *are* the schedule baseline. Figure 3.2 plots the same ES/EF values, colours launch-critical bars crimson, and shows remaining total float as a teal whisker to LF . Permitting, heritage, mobilisation and GSE run through August and September; October is the physical peak (stack 15 October, WDR 16–24 October, static fire 25–30 October); November is the flight-clearance tail.

Two Week 4 collisions are closed here. The charter dated pad integration 15 October in Objective 1 and stacking as M-4 on 28 October; the PEP locks M-4 to A-123 on **15 October 2026**, restoring the cold-flow window before hot fire. Handover “within five days of lift-off” was also dated M-8 at 22 November (seven days after a 15 November $T-0$); M-8 is now A-144 on **20 November**. Because the facility licence is already held, A-113 gates static fire, not stacking. Treating flight permits as a stacking predecessor was an initiation error.

Three gates sit on the baseline. Gate 1 is the PEP lock at A-111 on 2 October. Gate 2 is A-113 on 24 October: without those instruments A-133 cannot start, and that path is already $TF = 0$. Gate 3 is A-141 on 10 November. Crashing stacking without crashing A-113 therefore cannot pull $T-0$.

3.3 Schedule compression: crashing and fast-tracking

Kerzner (2017) and Project Management Institute (2021) distinguish fast-tracking (overlap of sequential work, usually at zero direct cost) from crashing (added resource on critical work, paid as a cost slope). The Board question is whether $T-0$ can move from 15 November to 10 November if R-14 is realised (ASA / CASA refuses the 15 November slot). Five days must come off a $TF = 0$ path that actually sets lift-off. After the A-133 merge that path is the tail A-134–A-143. Crashing A-123 by five days, or overlapping A-123 with A-124, saves **zero** days on $T-0$ while A-113 still finishes on 24 October. A-124 already runs in parallel with stacking.

The crash-cost slope on a crashed activity is

$$\text{Cost slope} = \frac{\Delta C}{\Delta T} = \frac{\text{crash cost} - \text{normal cost}}{\text{normal duration} - \text{crash duration}}. \quad (4)$$

Table 3.2 and Figure 3.3 compare three options that each claim five days. Only Option C is both CPM-valid and acceptable against the LRR hold point.

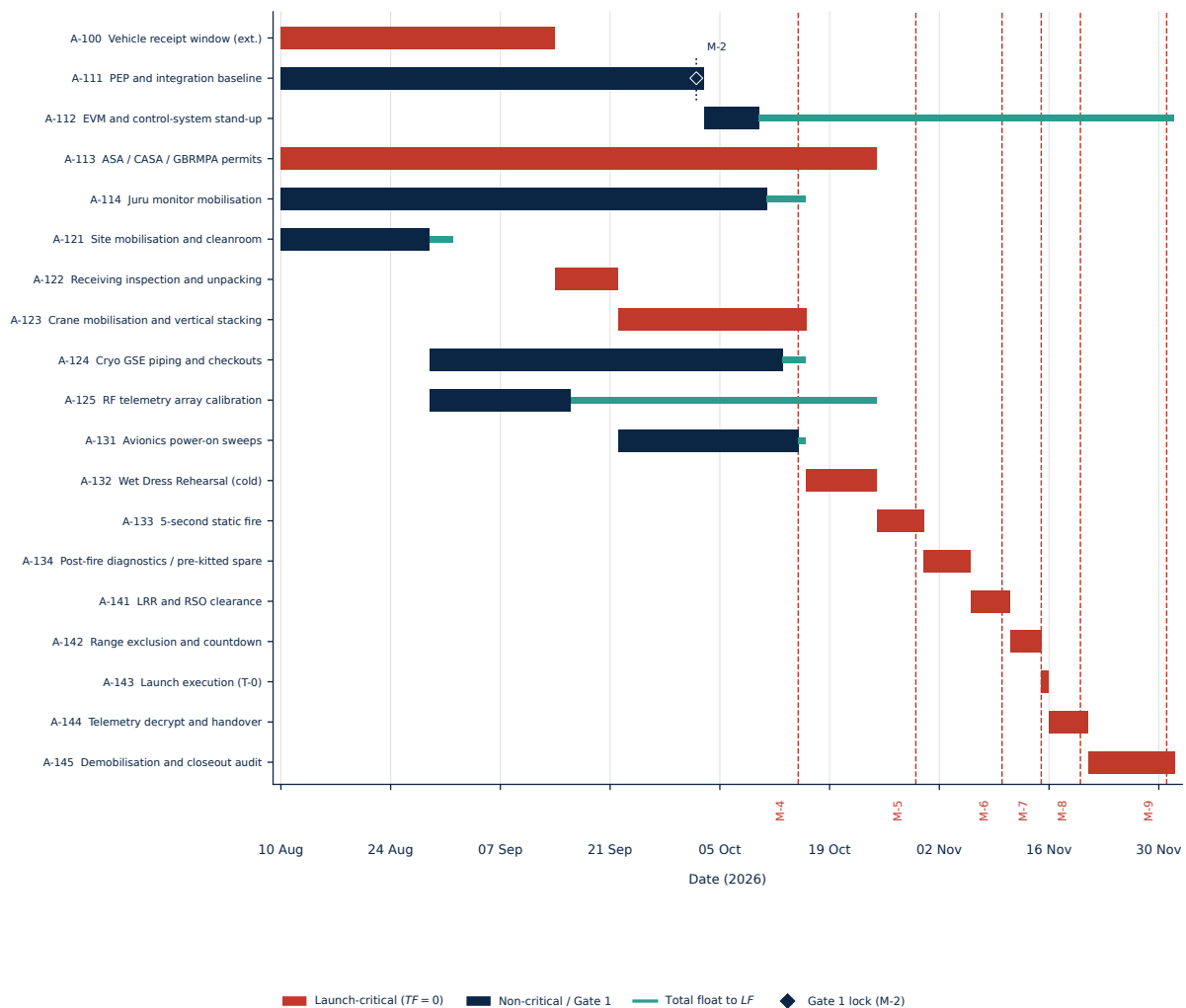


Figure 3.2: Baseline Gantt from 10 August to 1 December 2026, plotted from the same ES/EF values as Table 3.1. Crimson bars are launch-critical. Teal whiskers show total float to LF . Vertical markers are M-2 (navy, Gate 1) and M-4 through M-9. Option C is not drawn on this baseline.

Table 3.2: Schedule compression evaluation. All three options save five days on $T-0$ if they work as drawn. Baseline $T-0$ remains 15 November; Option C is a contingent draw on the \$81,050 reserve, not an addition to the \$1,500,000 base.

Option	Activities modified	Days	Added cost	Slope	Secondary risks	Verdict
A Pure fast-track	A-134 → A-141 SS+1 (LRR overlaps diagnostics)	5	\$0	\$0/d	LRR before diagnostic close-out; LRU rework; AS9100D hold-point failure	Rejected
B Pure crash	A-134 crashed 6 → 1 d	5	\$40,000	\$8,000/d	Fatigue; skipped thermal/electrical screens; recycle into A-141	Rejected
C Hybrid (if R-14 fires)	A-134 6 → 3 d (\$9,000); A-141 kept at 5 d; A-142 4 → 2 d (\$12,000)	5	\$21,000	\$4,200/d	R-15 residual is the 2 d sweep; A-141 hold-point duration kept	Contingent

Option A buys five days at zero direct cost by starting A-141 one day after A-134 starts. The review board would sit while diagnostics were still open—a hold-point failure under AS9100D (SAE International, 2016). Option A is rejected.

Option B compresses A-134 from six days to one. The slope is $\$40,000/5 = \$8,000$ per day. A 24-hour diagnostic war room on a vehicle that has just fired is the work most likely to skip thermal and electrical screens. Option B is rejected.

Option C splits the five days without shortening HP-4. Three days come off A-134 (\$9,000). Two days come off A-142 (\$12,000). Launch Readiness Review stays five days, finish-to-start after diagnostics. The weighted slope is $\$21,000/5 = \$4,200$ per day. If invoked, the dates become A-134 31 October–2 November, A-141 3–7 November, A-142 8–9 November, and A-143 on **10 November**. The marine-mammal buffer in A-142 shrinks from four days to two: that shorter sweep is the residual on R-15, which is why Option C is contingent rather than the baseline. The \$21,000 is drawn from the \$81,050 contingency only if R-14 is realised. It is not loaded into the \$1,500,000 base, and Figure 3.2 is not redrawn. Feeder floats do not tighten: A-131 stays at $TF = 1$, A-124 at $TF = 3$.

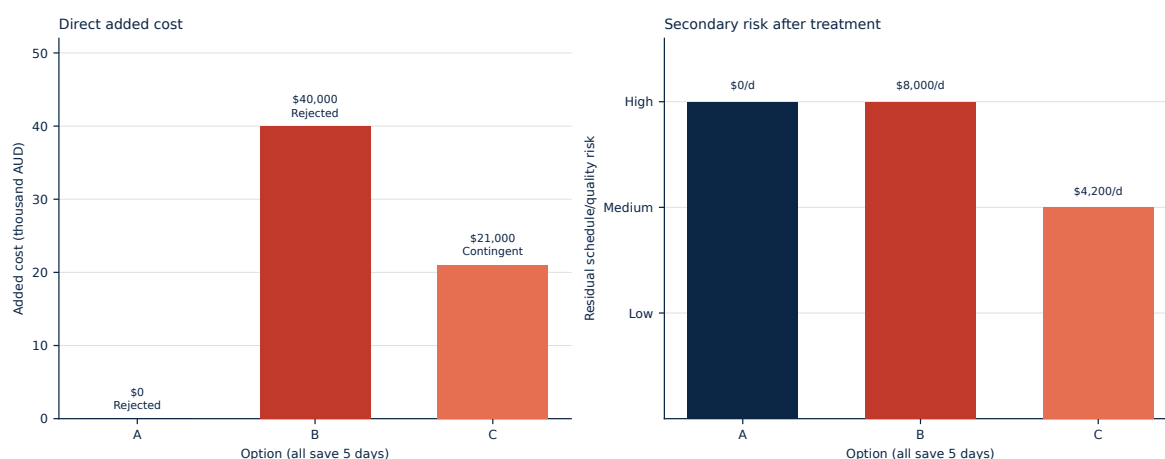


Figure 3.3: Compression trade-off for a five-day pull-in of $T-0$. Left: direct added cost. Right: residual schedule and quality risk after treatment. Option C is the only CPM-valid pull-in that keeps A-141 at five days and pays \$4,200 per day rather than \$8,000.

4 Cost management

The cost baseline is the WBS in Figure 2.1 priced at Level 3. Seventeen control accounts sum to a work estimate of \$1,500,000. Contingency of \$81,050 is sized from residual expected monetary value plus four named residual items, not from a round-number markup, so the authorised budget at completion is \$1,581,050.

4.1 Estimating basis and assumptions

The estimate is bottom-up at Level 3, with analogous plant and commodity rates beside the hours (Kerzner, 2017; Project Management Institute, 2021). Loaded 2026 Australian-dollar rates are used throughout: systems-engineering and project-management time at \$150/h, pad technicians at \$95/h, cryogenic and range-safety specialists at \$165/h, and surge technicians at \$100/h. The Manufacturing Award C10 ordinary hourly rate is \$29.45 from the first full pay period on or after 1 July 2026 (Fair Work Ombudsman, 2020). Charge-out is C10 multiplied by a named employment on-cost, then utilisation, overhead and a fee—the product is in the notes to Table 4.1. Those factors do not change the \$1,500,000 lines. The 100 t crawler for 23 m stacking is priced at \$4,200 per wet day, including operator, from published Queensland plant-hire schedules cross-checked against Rawlinsons (Rawlinsons, 2025).

Inclusions are the seventeen packages in Figure 2.1: mobilisation through demobilisation at Bowen. Factory manufacture, engine research and development, and freight to the receiving gate remain out of scope, so A-100 is a zero-dollar external driver rather than a cost account. The four-person surge crew (\$22,400) is inside WBS 1.3.3 as planned capacity. The \$21,000 hybrid compression cost in Section 3.3 is a contingent draw against the \$81,050 reserve if R-14 is realised; it is not in the \$1,500,000 performance measurement baseline.

The 7 August charter does not authorise field spend. Permitting and Juru work start on 10 August because those durations must finish before hot fire. Other August–September pad cost is a PEP choice, locked at Gate 1 (A-111, 2 October).

4.2 WBS cost baseline

Table 4.1 is the rollup. Pad logistics (1.2) is \$518,000 (34.5% of the work estimate), pad testing (1.3) \$414,000 (27.6%), governance (1.1) \$300,000 (20.0%), and flight/closeout (1.4) \$268,000 (17.9%).

Table 4.1: Bottom-up cost baseline by Level-3 WBS package. Totals equal the work estimate of \$1,500,000. A-100 is omitted because it is not a cost account.

WBS	Package	Amount (AUD)	Basis
1.1.1	PEP and integration	72,000	480 h PM @ \$150
1.1.2	Governance and EVM	48,000	320 h planner/cost @ \$150
1.1.3	ASA / CASA / GBRMPA	88,000	60-d ASA filing + CASA CRIS + GBRMPA
1.1.4	Juru monitors and heritage	92,000	Two monitors + survey
1.2.1	Receiving and cleanroom	68,000	Technician hours + consumables
1.2.2	Crane and 23 m stacking	218,000	8 lift + 16 standby d × \$4,200 = \$100,800 + riggers
1.2.3	Cryo GSE	180,000	Valves/piping + specialist crew
1.2.4	RF array calibration	52,000	Crew + analyser hire
1.3.1	Avionics power-on	58,000	Engineers @ \$150/h + EGSE
1.3.2	WDR (cold)	160,000	Includes bulk LOX 25 t and LN2 \$68,500
1.3.3	Static fire	148,000	Includes surge hire \$22,400
1.3.4	Post-fire / pre-kitted spare	48,000	Engineers + spare handling
1.4.1	LRR and RSO	38,000	Review board; payload witnesses
1.4.2	Range exclusion / marine	58,000	Patrol boats + NOTAM
1.4.3	Launch and telemetry capture	82,000	Launch-day operations
1.4.4	Decrypt and handover	24,000	Five-day engineering handover
1.4.5	Demobilisation and audit	66,000	Site restore + audit
Base		1,500,000	

Notes. Labour: MA000010 C10 ordinary \$29.45/h (ppc 1 July 2026) (Fair Work Ombudsman, 2020). Employment on-cost = super 12% + annual leave 7.69% + personal leave 3.85% + public holidays 4.23% + LSL 2.17% + 17.5% leave loading on annual leave + workers compensation (1.8% professional / 4.5% pad) + QLD payroll tax 4.75% (1.378 × professional, 1.405 × pad). Charge-out then divides by utilisation and applies overhead and a fee: engineer/PM $\$29.45 \times 1.378 / 0.70 \times 2.20 \times 1.176 = \150 ; technician $\$29.45 \times 1.405 / 0.85 \times 1.65 \times 1.183 = \95 ; specialist $\$29.45 \times 1.378 / 0.65 \times 2.40 \times 1.101 = \165 . Surge is C10 × 1.5 overtime × pad on-cost / 0.85 × 1.369 = \$100. Crane: 8 lift days and 16 non-lift standby days at the same wet \$4,200 (QLD 100 t crawler band about \$3,800–\$4,500/day), cross-checked against Rawlinsons (Rawlinsons, 2025). 1.1.3: ASA 60-day filing plus CASA cost-recovery plus GBRMPA (Australian Space Agency, 2025; Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019). LOX/LN2 \$68,500 is a bulk industrial-gas analogue inside 1.3.2. Surge \$22,400 = 4 × 7 d × 8 h × \$100 and sits in 1.3.3, not in contingency.

The largest package is crane mobilisation and stacking (1.2.2, \$218,000) because A-123 occupies 24 calendar days on the launch-critical path. The wet crawler is on hire for that whole window: 8 lift days and 16 non-lift days (weld, NDT, weather) at \$4,200, which is \$100,800, with the balance in rigger labour. R-06 residual EMV (\$2,160) is leftover wind probability beyond that planned standby; RES adds four further hire days (\$16,800) as duration uncertainty, not a second crane. Wet Dress Rehearsal (1.3.2) carries \$68,500 of bulk liquid oxygen and nitrogen

inside \$160,000. Static fire (1.3.3) carries the planned surge inside \$148,000. The seventeen lines sum to \$1,500,000 with no miscellaneous remainder. Package 1.2.1 remains one control account for A-121 and A-122.

4.3 Contingency

Contingency is not a 15% story. Table 4.2 adds a residual expected-monetary-value pool to four named residual items (Kerzner, 2017). Treatments already sit in the \$1,500,000 base (surge hire, dawn-lift window, dual-path recorders, pre-kitted spare), so the pool uses residual $P'_{\%} \times I_{\$}$, not inherent $P_{\%}$. Summing the eleven cost-bearing risks R-01, R-02, R-03, R-04, R-05, R-06, R-07, R-09, R-10, R-13 and R-14 gives \$37,920. The residual-uncertainty (RES) allowance is itemised in the same table and sums to \$43,130. Vehicle freight is out of scope and is not in RES. The two components sum to \$81,050, which is 5.40% of \$1,500,000, and the authorised BAC is therefore \$1,581,050.

Table 4.2: Contingency build-up. The 5.40% ratio is residual EMV plus four named residual items, not an input markup.

Component	Amount (AUD)	Rule
Σ residual EMV of eleven cost-bearing risks	37,920	Residual $P'_{\%} \times I_{\$}$; treatments already in the base
LOX/LN2 price variance (18% of \$68,500 gases)	12,330	In-scope commodity; not freight
Crane wind standby beyond R-06 EMV (4 d $\times$ \$4,200)	16,800	Extra hire days, not a second crane line
GN2/N2 purge cylinders and fittings	8,000	Named pad consumable
Cryogenic hose, seal and oxygen-clean kit	6,000	Named pad consumable
RES subtotal	43,130	Itemised; not a plug to 15%
Contingency reserve	81,050	5.40% of base
BAC	1,581,050	Base + contingency

Several register rows are kept out of the \$37,920 so that money is not counted twice. R-08 (pad-civil defect) is contract-transferred under the design-and-construct package and carries no Gilmour EMV. R-11, R-15, R-16, R-17 and R-18 are schedule or HSE exposures without a residual dollar draw. R-12 is the TestFlight 1 in-flight failure mode; the vehicle is factory product, so site-ops EMV is \$0. If R-14 is realised and Option C is invoked, the \$21,000 is drawn from the \$81,050 reserve. It is not added to the \$1,500,000 work baseline. There is no management reserve above BAC : unknown unknowns would require a board variation this plan does not authorise (Project Management Institute, 2021).

4.4 Time-phased budget and cash flow

The performance measurement baseline is the work estimate spread across the same dates as Table 3.1. Table 4.3 and Figure 4.1 use a uniform daily burn on each activity. October is the peak month because stacking, Wet Dress Rehearsal and the five-second static fire all sit there. December is a residual on 1 December, not a parking place for unused contingency.

Table 4.3: Time-phased planned value of the \$1,500,000 work baseline. Contingency is not monthly spend.

Month	Period PV (AUD)	Cumulative PV (AUD)
Aug 2026	155,951	155,951
Sep 2026	407,188	563,139
Oct 2026 (peak: stack, WDR, static fire)	628,861	1,192,000
Nov 2026	302,000	1,494,000
Dec 2026	6,000	1,500,000

The S-curve therefore terminates at \$1,500,000. Figure 4.1 draws $BAC = \$1,581,050$ as a horizontal authorisation line. Planned value at M-4 (15 October) is \$865,578—the same total used in Section 7.5. Option C's \$21,000 is excluded from every period in Table 4.3. If it is spent, it appears as actual cost charged to contingency, not as a rewrite of the S-curve.

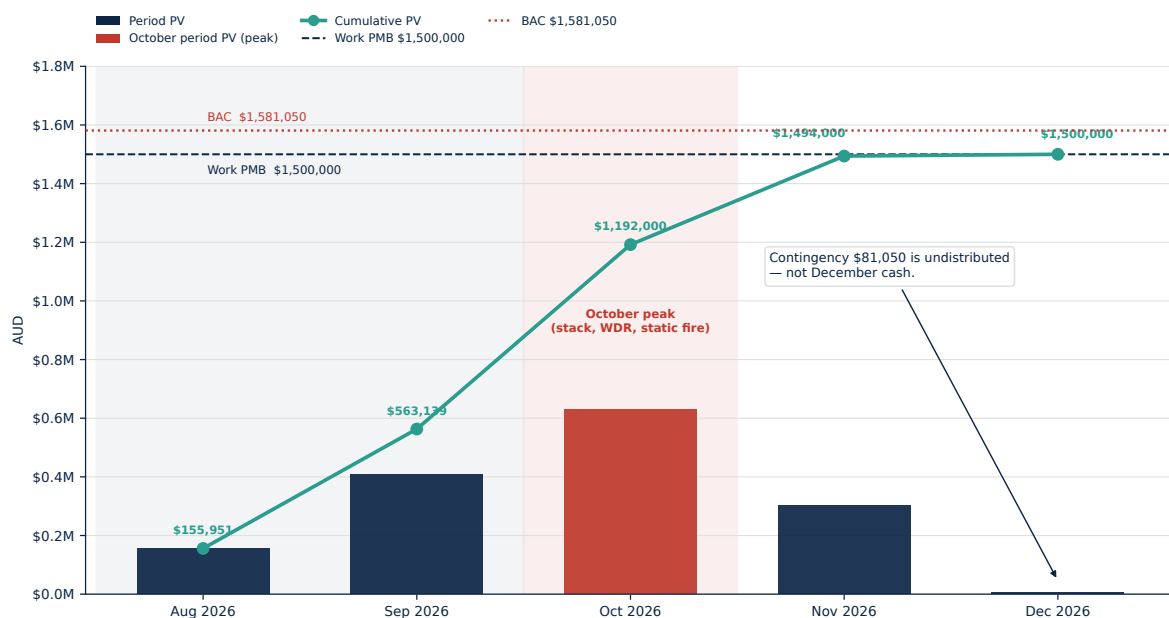


Figure 4.1: Cumulative planned-value S-curve for the Bowen campaign. Period bars are monthly PV; the teal line is cumulative PV and ends at the \$1,500,000 work baseline. The crimson dotted line is BAC (\$1,581,050) as an authorisation ceiling, not December cash. Contingency remains undistributed. Option C is not in the curve.

5 Resource management

Pad occupancy is a health-and-safety limit, not a roster convenience. The organic Bowen crew is eight certified technicians. The blast-zone, suit-air and access-control cap is twelve people. Eight is therefore the standing organisation; twelve is the occupancy ceiling that Figure 5.1 must never cross (Project Management Institute, 2021; Kerzner, 2017).

5.1 Roles and responsibilities

Accountability is split across two layers, with exactly one Accountable owner per Level-3 package in Table 5.1. The consultancy layer is Abdul Wahab Obeid, Hadi Alkhub, Ziyad Alahmari and Ilsa Hashmi. The field layer is the Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager. Consulted names must see the work product before the Accountable owner signs; they are not a second A.

The pattern matches the network. Hadi is Accountable for stacking because the dawn-lift window is a schedule control (R-06). Hot operations are Accountable to the Launch Director with the RSO consulted, except range exclusion, which the RSO owns. Closeout audit is Ziyad's, so quality is not marking its own homework.

Table 5.1: RACI for Level-3 packages. Exactly one Accountable owner per row. Field titles are as used on the pad; consultancy names are the Group 1 officers.

WBS	Package	A	R	C
1.1.1	PEP and integration	Abdul	Hadi	Ziyad, Ilsa
1.1.2	Governance and EVM	Ilsa	Hadi	Abdul
1.1.3	ASA / CASA / GBRMPA	Abdul	Hadi	Launch Director
1.1.4	Juru heritage	Juru Officer	Ilsa	Abdul, Ziyad
1.2.1	Receiving / cleanroom	GSE Lead	D&C mgr	Hadi, Ilsa
1.2.2	Crane and stacking	Hadi	GSE Lead	Launch Director, RSO
1.2.3	Cryo GSE	D&C mgr	GSE Lead	Ziyad
1.2.4	RF calibration	Avionics	Hadi	Launch Director
1.3.1	Avionics power-on	Avionics	GSE Lead	Hadi
1.3.2	WDR (cold)	GSE Lead	Juru Officer	Ziyad, LD, RSO
1.3.3	Static fire	Launch Director	GSE Lead	RSO, Ziyad, Avionics
1.3.4	Post-fire / pre-kitted spare	Avionics	GSE Lead	Ziyad
1.4.1	LRR and RSO	Launch Director	RSO	Abdul, Ziyad, Juru Officer, payload sponsor
1.4.2	Range exclusion	RSO	Launch Director	Juru Officer
1.4.3	Launch (T-0)	Launch Director	RSO, Avionics	Abdul
1.4.4	Telemetry handover	Avionics	Hadi	Ilsa
1.4.5	Demob and audit	Ziyad	D&C mgr	Ilsa, Abdul

5.2 Capacity planning

Figure 5.1 and Table 5.2 are the weekly simultaneous occupancy implied by Table 3.1. Each activity carries a named pad crew. Daily occupancy is the unique-head union of every crew whose bar is live that calendar day; the weekly bar is the maximum of those daily counts. The HSE cap of twelve is a simultaneous limit, not a weekly unique-head total. Two RF specialists belong only to A-125. Mitigated A-125 uses $ES-EF$ (29 August to 15 September), which is why weeks 3–6 sit two people above the unmitigated series. Unmitigated A-125 uses the late bar $LS-LF$ (7–24 October), so those two heads sit in weeks 9–11—on Wet Dress Rehearsal, not on static fire after 24 October. Four overtime heads are an unmitigated overlay on 16–30 October only. Together with late RF they take 16 October to fourteen: two above the cap. That is R-13 in numbers.

The baseline does two things, in that order. First, A-125 is placed at $ES-EF$ using its 39 days of total float. The move costs \$0 and does not change 15 November, because A-125 is a feeder into A-133, not a launch-critical bar. Second, four surge technicians are hired for seven days, 24–30 October, inside WBS 1.3.3 at \$100/h loaded: $4 \times 7 \times 8 \times \$100 = \$22,400$, already inside the \$1,500,000 work estimate (Fair Work Ombudsman, 2020). Those four replace the unmitigated overtime heads; they are not stacked on top of them. Mitigated 24–30 October is therefore eight named pad heads plus four surge, on the cap and never above it. Stacking ends 15 October and WDR starts 16 October, so there is no same-day overlap to invent. Option C, if invoked, shortens A-134 and A-142 and does not redraw Figure 5.1.

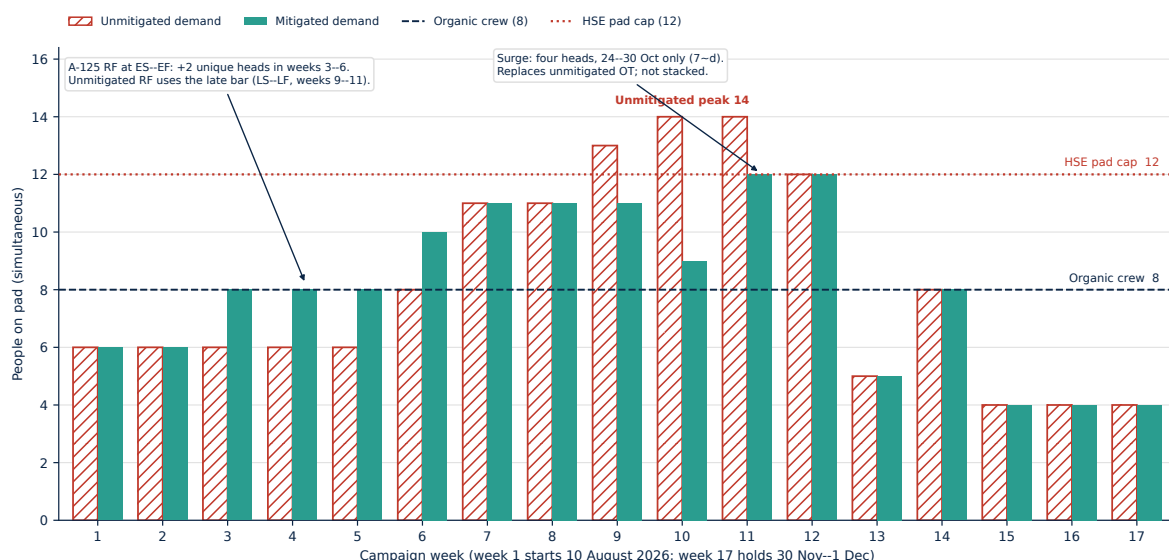


Figure 5.1: Weekly pad demand versus capacity, from Table 5.2. Each bar is the maximum simultaneous unique-head count that week. Hashed crimson bars are the unmitigated series (peak 14 on 16–24 October): Wet Dress Rehearsal plus A-125 on its late bar plus four overtime heads. Teal bars are the mitigated baseline (peak 12): RF at $ES-EF$ in weeks 3–6, and four surge technicians on 24–30 October only. The navy dashed line is the organic crew of eight; the crimson dotted line is the HSE occupancy cap of twelve. Eight is not a hard pad limit.

Table 5.2: Weekly simultaneous pad occupancy, from activity $\times$ named crew. Each weekly total is the maximum unique-head count on any calendar day that week (HSE cap is simultaneous, not weekly unique). Mitigated A-125 uses $ES-EF$; unmitigated uses the late bar $LS-LF$. Surge is four heads on 24–30 Oct only (7 days). Unmitigated overtime is four heads on WDR and static fire (16–30 Oct). Base, RF, surge and OT are disjoint name sets, so they add on the peak day.

Week	Mit. day	Base	RF	Surge	Mit.	Unmit. day	Base	RF	OT	Unmit.
1	10 Aug	6	0	0	6	10 Aug	6	0	0	6
2	17 Aug	6	0	0	6	17 Aug	6	0	0	6
3	29 Aug	6	2	0	8	24 Aug	6	0	0	6
4	31 Aug	6	2	0	8	31 Aug	6	0	0	6
5	7 Sep	6	2	0	8	7 Sep	6	0	0	6
6	14 Sep	8	2	0	10	14 Sep	8	0	0	8
7	22 Sep	11	0	0	11	22 Sep	11	0	0	11
8	28 Sep	11	0	0	11	28 Sep	11	0	0	11
9	5 Oct	11	0	0	11	7 Oct	11	2	0	13
10	12 Oct	9	0	0	9	16 Oct	8	2	4	14
11	24 Oct	8	0	4	12	19 Oct	8	2	4	14
12	26 Oct	8	0	4	12	26 Oct	8	0	4	12
13	6 Nov	5	0	0	5	6 Nov	5	0	0	5
14	15 Nov	8	0	0	8	15 Nov	8	0	0	8
15	21 Nov	4	0	0	4	21 Nov	4	0	0	4
16	23 Nov	4	0	0	4	23 Nov	4	0	0	4
17	30 Nov	4	0	0	4	30 Nov	4	0	0	4

5.3 Conflict resolution and fallback

Resource conflicts escalate on the work, not on personalities. Tier 1 field leads have four hours to level, stagger or borrow from float without spending money. Tier 2 is the campaign PM: twelve hours to choose overtime, hire or resequence, with the cost consequence going to Ilsa so that Table 4.1 stays the book of record. Tier 3 is the Executive Board, with a 24-hour SLA, for any decision that consumes contingency or moves M-5 (30 October) or M-7 (15 November). Those gates are the same dates as Section 3; they are not a second calendar.

Team-member fallback is separate and is the A1 defect closed here. If a member is non-responsive for more than 48 hours, the Lead Planner audits the artefacts within 12 hours, the PM issues a WBS-reassignment addendum, and the Buddycheck PAF is locked against Teams and version history so that contribution evidence cannot be rewritten after the fact. That procedure reallocates packages in Table 5.1; it does not authorise a silent change to $T-0$ or to BAC .

6 Risk management

Campaign risk is owned by Ziyad Alahmari (Risk and Quality Lead) and is scored on this pad, not as generic weather or budget-overrun rows. Eighteen cause–event–effect statements in Table 6.1 size the \$81,050 contingency already reported in Table 4.2, name the float that absorbs non-critical delay, and fix the owners who treat each item.

6.1 Risk approach and scales

The process follows ISO 31000 identify–analyse–evaluate–treat–monitor (International Organization for Standardization, 2018). Expected monetary value is calculated as in Kerzner (2017): money is never read from the 1–5 heat-map scores. Inherent $EMV = P_{\%} \times I_{\$}$ uses the dollar column in Table 6.1. Contingency uses residual $EMV = P'_{\%} \times I_{\$}$ because the named treatments already sit in the \$1,500,000 base. Probability bands are $P = 1$ below 5% (3% in EMV), $P = 2$ from 5–20% (12%), $P = 3$ from 21–40% (30%), $P = 4$ from 41–60% (50%), and $P = 5$ above 60% (70%). Schedule impact I_S is 1 (< 2 d), 2 (2–5 d), 3 (6–10 d), 4 (11–15 d) and 5 (> 15 d on a $TF = 0$ path). Cost impact I_C uses a stated dollar scale: 1 (< \$15k), 2 (\$15k–\$39,999), 3 (\$40k–\$79,999), 4 (\$80k–\$149,999), 5 (\$150k or more). The matrix coordinate I is the maximum of those two bands (Project Management Institute, 2021). Residual P falls only where the named response actually bites. Residual I falls only where the effect itself is smaller; a 12–14 day recycle remains $I = 4$ even if it is less likely. Ziyad re-scores the register at Gate 1 (2 October), at M-4 (15 October) and at M-5 (30 October), in the same cadence as the EVM dashboard in Section 7.5.

6.2 Integrated risk register

Table 6.1 is the full set. Cause, event and effect are three clauses. Owners match the RACI in Table 5.1: Ziyad with the GSE Lead on static-fire abort (R-01) and township acoustics (R-17); the Launch Director on footprint, stage-separation and the in-flight TestFlight 1 mode (R-02, R-09, R-12); Ilsa Hashmi on LOX logistics, occupancy and the Juru commercial interface (R-10, R-13, R-04 with the Juru Cultural Officer); Hadi Alkhub on the dawn-lift window and countdown software (R-06, R-18); Abdul Wahab Obeid on permits and LRR deadlock (R-14, R-16). Figure 6.1 plots every inherent point and the residual after treatment. R-02, R-09 and R-12 share high-impact cells and are offset.

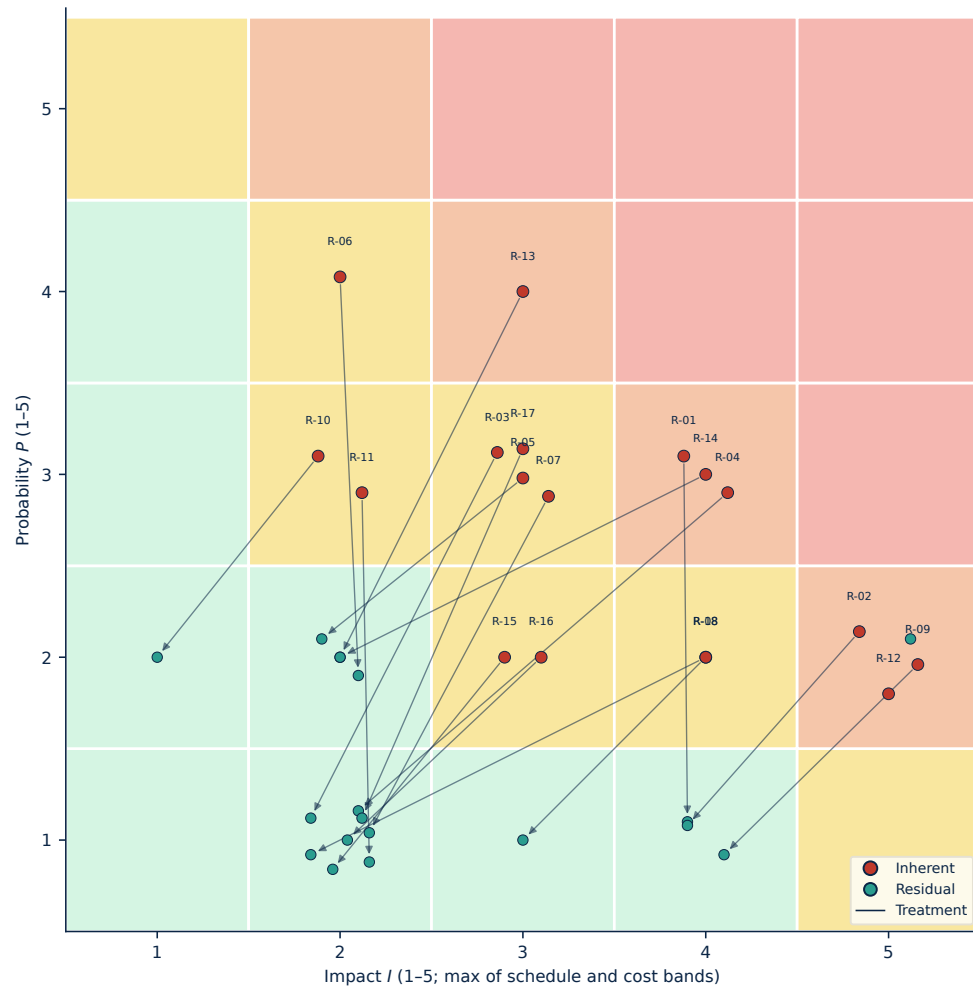


Figure 6.1: Inherent (crimson) to residual (teal) movement for all eighteen register items. P is vertical and I is horizontal. Most treatment arrows run down and left. R-01 keeps $I = 4$ (a recycle is still 12–14 days). R-12 keeps $I = 5$ (in-flight loss still ends heritage). R-02, R-09 and R-12 share high-impact cells and are offset.

Table 6.1: Integrated campaign risk register. $P \times I$ uses the 1–5 scores. The EMV column is residual $P'_{\%} \times I_{\$}$, which feeds Table 4.2. A dash means the row is not in the \$37,920 pool.

ID	Anchor	Cause → event → effect	P	I	$P \times I$	Owner	Response	P'	I'	EMV
R-01	A-133 / 1.3.3	Residual pump insulation → static-fire auto-abort → 12–14 d recycle / \$90k	3	4	12	Ziyad / GSE Lead	Thermal soak; pre-kitted LRU spare; 48 h swap. Design defects return to R&D	1	4	2,700
R-02	A-143 / 1.4.3	Trajectory energy toward park boundary → debris in GBRMPA zone → statutory halt	2	5	10	Launch Director	South-biased LRR footprint; joint GBRMPA protocol	1	4	5,400
R-03	A-125 / A-143	RF interference at Bowen array → telemetry <98.5% → heritage fail	3	3	9	Avionics Lead	Dual-path recorders; A-125 before hot fire; mobile backup	1	2	1,500
R-04	A-114 / A-143	Exclusion without stop-work → Juru complaint → range withheld	3	4	12	Ilsa / Juru Officer	Embedded monitors with stop-work; cultural HP before A-132	1	2	1,200
R-05	A-124 / A-132	Inadequate purge before LOX → frozen/cracked valve → WDR abort ~8 d	3	3	9	GSE Lead	LN2 pre-chill; spare valve kit; helium leak test HP-2	2	2	6,600
R-06	A-123 / 1.2.2	Abbot Point crosswind >20 kt at vertical mate → lift abort → crane standby	4	2	8	Hadi / Stack Conductor	Dawn-lift window; standby already inside 1.2.2	2	2	2,160
R-07	A-124 / D&C	D&C GSE IFC lags propulsion ICD → pad mismatch → rework	3	3	9	GSE Lead / PM	ICD freeze gate; weekly integration; D&C LDs	1	2	2,100
R-08	1.2.3 / D&C	Latent pad-civil defect → hold-down failure → hot-ops stop	2	4	8	D&C mgr (A: PM)	NDT HP-1; defect liability remains with D&C	1	2	0*

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause → event → effect	<i>P</i>	<i>I</i>	<i>P</i> × <i>I</i>	Owner	Response	<i>P'</i>	<i>I'</i>	EMV
R-09	A-143	Stage-sep charge timing error → debris outside footprint → inquiry / halt	2	5	10	Launch Director	Dual-string sep inhibit; wind hold toward park	1	4	4,500
R-10	A-132 / 1.3.2	Single 800 km LOX tanker delay → WDR cannot load → 2–5 d slip	3	2	6	Ilsa	72 h tanker confirm; second-tanker backup; RES on price	2	1	2,760
R-11	A-131	Flight-computer vs pad EGSE skew → power-on abort	3	2	6	Avionics Specialist	ICD freeze; cleanroom EGSE dry run before stack	1	2	—
R-12	A-143	Two-motor pump fault after <i>T</i> -0 → thrust loss as at <i>T</i> +14 s → heritage fail	2	5	10	Launch Director	HP-3 screens ground abort; in-flight mode accepted (<i>P</i> stays 2); no on-pad redesign	2	5	—
R-13	A-132 / A-133	Peak 14 vs cap 12 → detank error → recycle / injury	4	3	12	Ilsa	A-125 moved to Sep; four surge techs in base; 10 h stagger	2	2	3,600
R-14	A-113 / A-143	Late ASA filing or CASA NOTAM → 15 Nov slot refused	3	4	12	Abdul / Regulatory	File 60 d prior (A1 Keep Satisfied); weekly ASA/CASA liaison; Option C backup	2	2	5,400
R-15	A-142	Dugong in exclusion box → sweep extension	2	3	6	RSO	Marine-mammal protocol; A-142 holds 4 d on the baseline. If Option C fires, residual is the 2 d sweep	1	2	—

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause → event → effect	<i>P</i>	<i>I</i>	<i>P</i> × <i>I</i>	Owner	Response	<i>P'</i>	<i>I'</i>	EMV
R-16	A-141	RSO vs Launch Director deadlock at LRR → missed 10 Nov gate	2	3	6	Abdul / Launch Director	Named DoA; escalate 24 h before A-141 <i>LF</i>	1	2	—
R-17	A-133	Static-fire acoustic >135 dB at Bowen → complaint / hold	3	3	9	Ziyad / HSE	Water deluge; 1.5 km exclusion; community notice 14 d prior	1	2	—
R-18	A-142 / A-143	FTS/countdown abort at <i>T</i> -10 → recycle inside window	2	4	8	Hadi / Avionics	Two countdown rehearsals; abort-to-recycle procedure	1	3	—

*R-08 is contract-transferred (EMV \$0 to Gilmour). R-11, R-12, R-15–R-18 are schedule, HSE or out-of-scope vehicle heritage; R-12 has no site-ops dollar draw. Residual EMV identities: R-01 $0.03 \times 90,000$; R-02 $0.03 \times 180,000$; R-03 $0.03 \times 50,000$; R-04 $0.03 \times 40,000$; R-05 $0.12 \times 55,000$; R-06 $0.12 \times 18,000$; R-07 $0.03 \times 70,000$; R-09 $0.03 \times 150,000$; R-10 $0.12 \times 23,000$; R-13 $0.12 \times 30,000$; R-14 $0.12 \times 45,000$. Sum \$37,920.

6.3 Top risks, contingency, float and delivery model

Three rows dominate board attention because they sit on the launch-critical path and can stop $T-0$ or close the heritage case.

R-01 is a residual pump-insulation defect on A-133: auto-abort, a 12–14 day recycle, about \$90k of consumables. Ziyad and the GSE Lead treat it with thermal-soak instrumentation, a pre-kitted LRU spare and a 48-hour swap. Design defects still return to R&D. The score falls from (3, 4) to (1, 4): probability drops, but residual impact stays 4 because a recycle is still 12–14 days. Residual EMV is \$2,700 ($3\% \times \$90k$), not the inherent \$27,000, because the soak, spare and swap already sit in the work baseline.

R-12 is the TestFlight 1 in-flight mode: thrust loss in two first-stage hybrid motors after $T-0$, as at $T+14$ s, ending the heritage objective. The Launch Director owns HP-3 as the static-fire screen for a ground abort. HP-3 does not treat an event after $T-0$, so residual probability stays $P' = 2$. Residual impact stays 5 because an in-flight loss still closes heritage. There is no on-pad redesign and no site-ops dollar draw (EMV \$0).

R-14 is an ASA flight-authorisation or CASA NOTAM refusing the 15 November slot. Abdul files sixty days prior, as the charter scored ASA Keep Satisfied, and holds a weekly ASA/CASA liaison. Crashing A-123 does not treat R-14: permits already finish on 24 October with $TF = 0$. Option C is the schedule backup if this row fires. It is not justified by an invented 10–12 November payload geometry.

R-09 (stage-separation timing) and R-02 (footprint toward the park) remain high-impact register rows on the same LRR protocol. They are not the three board headlines.

Those eleven cost-bearing rows sum to \$37,920 of residual EMV. Ilsa's itemised RES allowance of \$43,130 covers in-scope LOX price variance, crane wind standby, GN2/N2 purge stock and a cryogenic hose/seal kit. Together they are the \$81,050 in Table 4.2 (5.40% of base, not a 15% markup). Vehicle freight stays out of RES. R-06 standby dollars already sit inside WBS 1.2.2, which is why that residual EMV is \$2,160 rather than a second crane line.

Float, not money, is the treatment for several feeders. A-125's 39 days let Ilsa park RF in weeks 3–6 so R-13 never meets an occupancy of 14. A-124's three days and A-114's five days absorb GSE and heritage delay before A-132 without moving 15 November. R-08 is transferred after HP-1 NDT. If R-14 fires, Abdul takes Option C to the Board and Ilsa draws \$21,000 from contingency. That draw is not in the \$1,500,000 work baseline, it does not move the M-7 diamond on Figure 3.2 until the Board says so, and the residual on R-15 becomes the two-day marine sweep.

7 Delivery and control

This section converts the baselines in Sections 3–6 into decision rights, a pad-specific procurement choice, hold points, Bowen HSE, and an earned-value framework that will be read at M-4.

It is a control design, not a progress report: the campaign has not been executed.

7.1 Governance and reporting

Governance is the organisation that will actually stop work, not a generic pyramid (Project Management Institute, 2021; Kerzner, 2017). The Gilmour Executive Board is the sponsor. Abdul Wahab Obeid is Campaign Project Manager and directs the three consultancy leads in Table 5.1. Field titles on Figure 7.1 match that table: Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager.

Delegation is numeric. Abdul may approve a variance of at most \$20,000 *and* at most 48 hours, and only if M-5 (30 October, A-133) and M-7 (15 November, A-143) do not move. Anything else—a contingency draw, Option C's \$21,000, a hold-point waiver, a *T*-0 change—goes to the Board within 24 hours (Kerzner, 2017). R-16 (RSO versus Launch Director deadlock) escalates 24 hours before A-141's late finish of 10 Nov, not inside the review room. The \$20,000 ceiling is below the dollar *impact* of the hot-fire rows that would recycle the pad (R-01 \$90,000; R-07 \$70,000). Residual EMVs on those rows are \$2,700 and \$2,100 because treatments already sit in the work baseline; a genuine recycle still cannot be buried in PM discretion.

Cadence matches the physical work. Field leads hold a daily pad standup. After Gate 1 (A-111, 2 Oct) Ilsa publishes a weekly EVM dashboard from A-112. Abdul takes a fortnightly pack to the sponsor. Daily standup does not rewrite Table 3.1. A-112 has 53 days of float so the control system does not compete with stacking for pad occupancy.

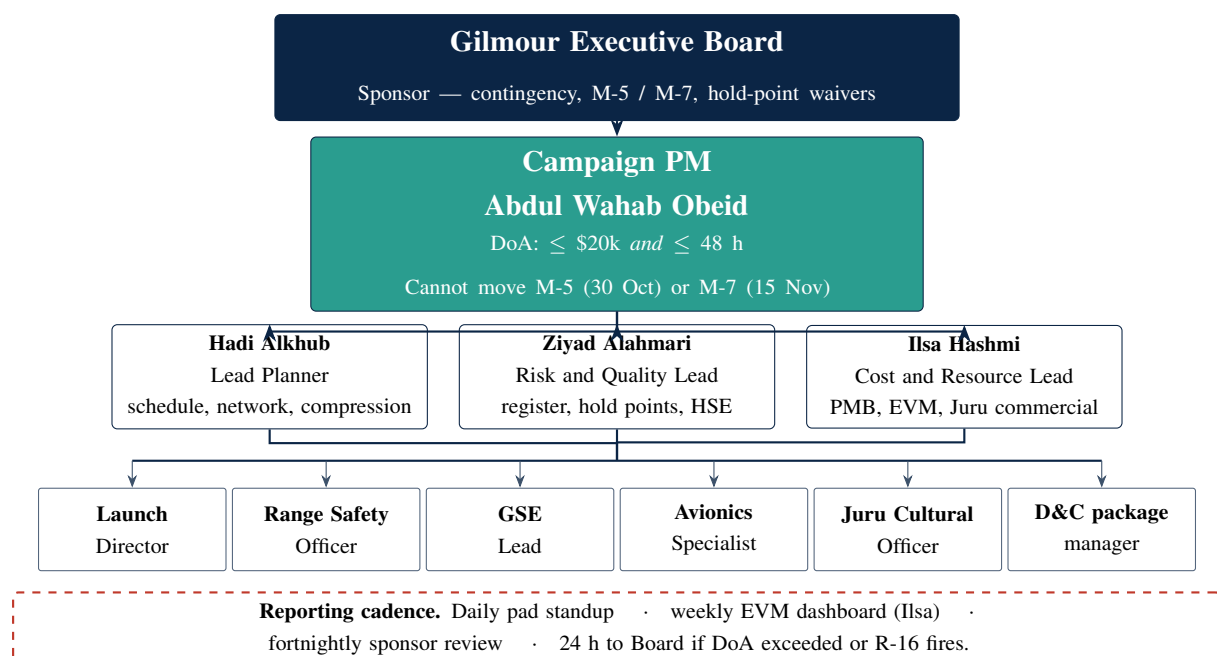


Figure 7.1: Campaign governance for Bowen pad operations. Named roles, a numeric DoA on the PM, and the reporting cadence. Field titles match Table 5.1. The Board, not the PM, moves M-5 or M-7 or draws the \$81,050 reserve.

7.2 Delivery model and procurement

The chosen delivery model is a managing-contractor / design-and-construct hybrid (Turner, 2014; Project Management Institute, 2021). Gilmour retains propulsion intellectual property, vehicle integration and the Australian launch licence. Civil works and cryogenic GSE packages are let as design-and-construct, with an interface-control freeze and liquidated damages. That split is the contractual answer to R-07 and R-08: an ICD lag is an LD event on the contractor, and a latent pad-civil defect stays with the D&C package after HP-1 NDT (R-08 EMV \$0 to Gilmour). Abdul stays Accountable on that row; the package manager stays Responsible.

Turnkey EPC was considered and rejected. After TestFlight 1, handing the propulsion stack and the launch authorisation to a single EPC would put flight-critical intellectual property and statutory accountability in a contractor who does not hold the licence (Gilmour Space Technologies, 2025; Australian Space Agency, 2025). Two Turner tests decide the model (Turner, 2014). Scope certainty is high on site operations—seventeen Level-3 packages, locked dates, a \$1,500,000 work estimate—and low on the vehicle. Client capability is the other test: Gilmour can integrate a launch vehicle; it should not pour pad grout or run a 100 t crawler as a core competence. The charter’s weighted screen (Strategic Fit 40%, Technical Feasibility 30%, Schedule 30%) reaches the same rejection. Alliance contracting was the second rejected alternative: a four-month campaign cannot carry pain-share overhead, and GBRMPA and CASA duties are not commercially shareable (Great Barrier Reef Marine Park Authority, 2019; Civil Aviation Safety Authority, 2024).

7.3 Quality

Quality is specified against AS9100D and ISO 9001:2015 (SAE International, 2016; International Organization for Standardization, 2015). Four hold points, each tied to a register row, are listed in Table 7.1. HP-1 is pad tie-down weld NDT on A-123 (R-08). HP-2 is the GSE helium leak test on A-124 (R-05). HP-3 is the pre-static-fire ignition interlock on A-133 (R-01). HP-4 is LRR sign-off on A-141: Launch Director, RSO and Juru Officer review with Abdul and Ziyad; the payload sponsor witnesses and does not sign. Work does not proceed past a hold without Ziyad’s release or a Board waiver, and a Board waiver is outside Abdul’s DoA.

Table 7.1: Quality hold points for the Bowen campaign. Each hold is a named artefact, not a meeting. Measurement rules in Section 7.5 follow the same IDs.

Hold	Activity	Risk	Artefact	Release
HP-1	A-123 (15 Oct)	R-08	Pad tie-down weld NDT	D&C mgr / Ziyad
HP-2	A-124 (12 Oct)	R-05	GSE helium leak test	GSE Lead / Ziyad
HP-3	A-133 (30 Oct)	R-01	Pre-static-fire ignition interlock	GSE Lead / Launch Director / Ziyad
HP-4	A-141 (10 Nov)	R-16	LRR sign-off (payload sponsor witnesses)	Launch Director / RSO / Juru Officer / Abdul / Ziyad

Measurement follows the holds and is locked in YAML. Discrete tests and flight artefacts—A-132, A-133, A-141, A-143—are 0/100. A-113 is elapsed-day percent-complete because the work is a 76-day wait, not a binary test. Installations—A-123 and A-124—are milestone percent-complete tied to HP-1 and HP-2. LRR waits on A-134 rather than on A-133: HP-4 cannot be signed on incomplete post-fire data, which is why Option A was rejected (SAE International, 2016).

7.4 Health, safety and environment

HSE is this pad under the *Work Health and Safety Act 2011* (Qld) (Queensland Government, 2011). Liquid oxygen is the process hazard: LOX PPE, oxygen-clean tools, and exclusion during load and detank on A-132 and A-133. Blast occupancy is a 1.5 km zone and twelve people—the same cap on Figure 5.1. Acoustic treatment is a water deluge so static-fire level at Bowen township stays at or below 135 dB (R-17), with community notice fourteen days prior. Juru artefact buffers and stop-work authority (R-04) are HSE controls: A-114 feeds A-132 as a cultural hold. Finish-line HSE remains as chartered: zero lost-time injuries, zero GBRMPA breaches, and zero Juru heritage breaches, with a five-second static fire of the four first-stage hybrid engines and handover of at least 98.5% of telemetry. Marine-mammal protocol on A-142 (R-15) is owned by the RSO (Great Barrier Reef Marine Park Authority, 2019). If R-14 fires, Option C shortens A-134 and A-142 and keeps A-141 at five days so HP-4 is not crashed. Occupancy never exceeds twelve on the mitigated histogram: R-13 is already treated in the \$1,500,000 base by moving A-125 and hiring four surge technicians.

7.5 Earned value management

Earned value is the instrument that will tell the Board whether the \$1,581,050 authorisation is still intact (Project Management Institute, 2019; Fleming and Koppelman, 2016; Kerzner, 2017). It is a framework. The campaign has not started, so there is no fourteen-week EV/AC history to

plot.

Distributed planned value is \$1,500,000, identical to Table 4.1 and to the S-curve in Figure 4.1. Contingency of \$81,050 is undistributed budget: it is not December cash and it is not monthly PV. Authorised BAC is \$1,581,050. Option C's \$21,000 is not in the work PMB; if R-14 fires and the Board invokes it, the draw appears as actual cost charged to contingency.

SV and CV are currency, never days:

$$SV = EV - PV, \quad CV = EV - AC, \quad SPI = EV/PV, \quad CPI = EV/AC. \quad (5)$$

In-flight estimate at completion on the work baseline is

$$EAC = AC + (BAC_{\text{work}} - EV)/CPI, \quad (6)$$

with $BAC_{\text{work}} = \$1,500,000$. The same CPI is applied to the authorisation ceiling as $BAC \times AC/EV$, reported against \$1,581,050 (Project Management Institute, 2019). Finish-line rule: final $AC \leq \$1,581,050$. A Green CPI of 0.96 in October is allowed; finishing above BAC is not.

Thresholds that trigger action, used only in this section, are:

- **Green** — $0.95 \leq CPI, SPI \leq 1.05$: dashboard only.
- **Amber** — $0.90 \leq \text{index} < 0.95$: Exception Report and a seven-day recovery plan within 48 hours.
- **Red** — $\text{index} < 0.90$, or $T-0$ slip > 2 days: contingency freeze and Board intervention. SV is not written in days.

The only dashboard numbers permitted in this PEP are the two illustrative readings at M-4 (15 October 2026) in Table 7.2 and Figure 7.2. Planned value at that date is \$865,578. The planned label is HP-1 signed (A-123 100%), HP-2 signed, A-113 elapsed-day percent-complete. On-plan books EV equal to PV (\$865,578) and AC \$901,000, so $CPI = 865,578/901,000 = 0.96$ and $SPI = 865,578/865,578 = 1.00$ —Green. Even so,

$$EAC = 901,000 \times 1,500,000/865,578 = \$1,561,384,$$

which is identical to $AC + (BAC_{\text{work}} - EV)/(EV/AC)$. Authorisation

$$BAC \times AC/EV = 1,581,050 \times 901,000/865,578 = \$1,645,751$$

sits above the \$1,581,050 ceiling. Green at M-4 still carries a recovery obligation toward the finish-line rule.

The late-stack illustration holds PV at \$865,578 but drops EV to \$647,578 and AC to \$870,000 because HP-1 is unsigned. A-123 is milestone 0/100 tied to that hold, so the \$218,000

control account contributes \$0 of EV—the whole package is withheld, not a partial-hours claim. Then $CPI = 647,578/870,000 = 0.74$ and $SPI = 647,578/865,578 = 0.75$ —Red. Ilsa issues the Exception Report within 48 hours and freezes contingency. If R-14 has fired, the seven-day recovery invokes Option C and crashes A-134 and A-142—not A-141. Work EAC is $870,000 \times 1,500,000/647,578 = \$2,015,201$. $BAC \times AC/EV = 1,581,050 \times 870,000/647,578 = \$2,124,089$. Those two points are a worked example at one physical gate, not a time series and not a claim that October has already been worked.

Table 7.2: Illustrative earned-value status at M-4 (15 October 2026). The campaign has not been executed. PV is the locked work-PMB value at stacking complete. Indices use the reported two-decimal CPI/SPI from EV/AC and EV/PV .

Status at M-4	PV	EV	AC	CPI	SPI	Action
HP-1 signed (A-123 100Illustrative on-plan	865,578	865,578	901,000	0.96	1.00	Green — weekly dashboard
Illustrative late stack	865,578	647,578	870,000	0.74	0.75	Red — 48 h exception report; contingency freeze

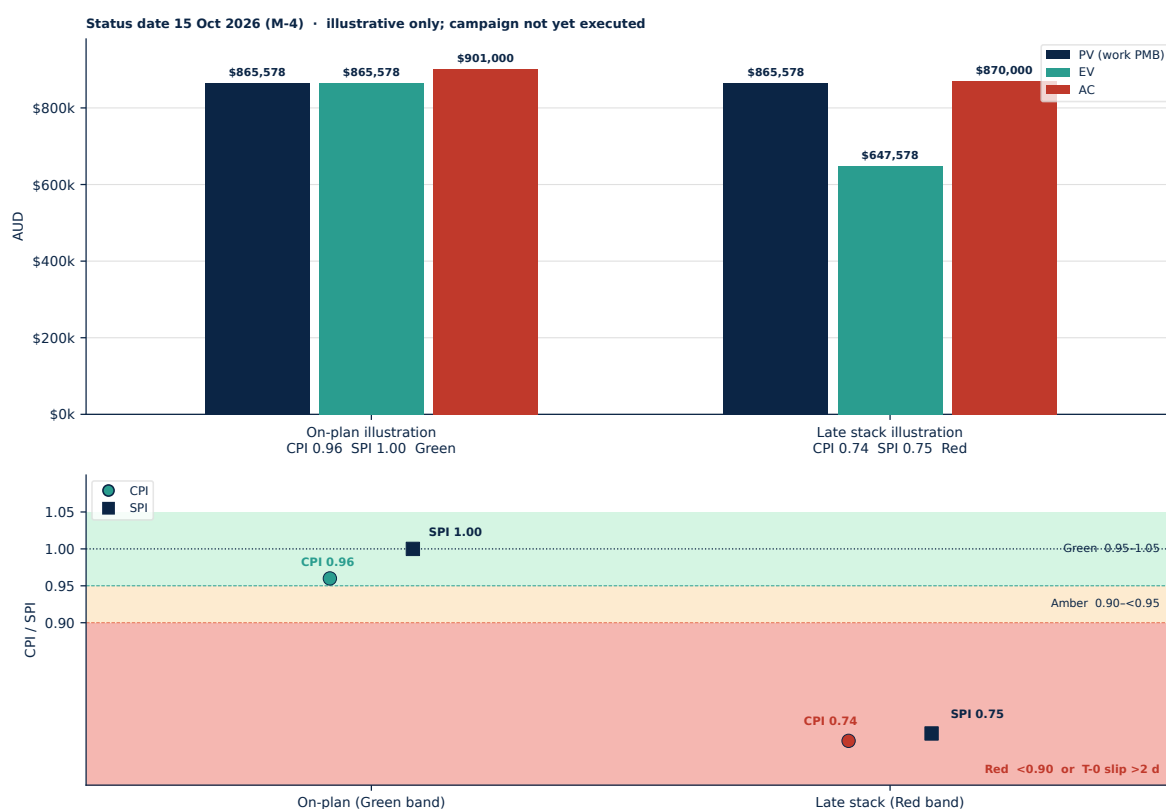


Figure 7.2: Illustrative status at M-4; campaign not yet executed. Upper panel: PV, EV and AC at the single status date 15 October 2026 for the on-plan and late-stack cases. Lower panel: Green / Amber / Red CPI–SPI bands, with the on-plan pair (0.96, 1.00) in Green and the late-stack pair (0.74, 0.75) in Red. Red is also triggered by $T-0$ slip greater than 2 days. No weekly EV/AC history is implied.

8 Integration and professionalism

This PEP is one campaign described eight times. Scope, WBS, network, dollars, occupancy, register, hold points and earned-value bands name the same Bowen pad, the same seventeen cost accounts, and the same 15 November $T-0$. Where a view disagrees with Table 8.1, `data/pep_baseline.yaml` is the book of record.

8.1 Document consistency

The identities are already in the body and are collected in Table 8.1. The work estimate is \$1,500,000. Contingency is \$81,050 (residual EMV \$37,920 plus RES \$43,130; 5.40% of base, not a 15% markup). Authorised BAC is \$1,581,050. M-4 is A-123 on 15 October; M-5 is A-133 on 30 October after cold WDR; M-6 is A-141 after diagnostics; M-7 is A-143 on 15 November; M-8 is A-144 on 20 November. Option C's 10 November lift-off is a contingent \$21,000 draw if R-14 fires. Occupancy peaks at twelve simultaneous unique heads after A-125 is parked on float and \$22,400 of surge labour is placed inside WBS 1.3.3 for 24–30 October only.

Table 8.1 is that thread in one view. SV is currency, never days. The S-curve is not forced to \$1,581,050. Eight people are the organic crew, not a hard pad limit.

Table 8.1: Golden-thread reconciliation. Every view in this PEP is required to show the value in the middle column; the right-hand column is where the marker should look.

Thread	Locked value	Views that must agree
Work PMB	\$1,500,000	Sec. 1; Sec. 2.4; Table 4.1; Table 4.3 (ends here); Fig. 4.1; Sec. 7.5 distributed <i>PV</i>
Contingency	\$81,050 = EMV \$37,920 + RES \$43,130	Table 4.2; Table 6.1 dollar column; 5.40% of base, not a 15% markup
Authorised <i>BAC</i>	\$1,581,050	Finish-line $AC \leq BAC$ in Sec. 1, Sec. 2.4 and Sec. 7.5
M-4 stacking	15 Oct 2026 = A-123 <i>EF</i>	Table 2.1; Table 3.1; Fig. 3.2; Fig. 7.2 status date
M-5 static fire	30 Oct 2026 = A-133 <i>EF</i>	Cold WDR A-132 then hot fire A-133; HP-3 on A-133
M-6 LRR	10 Nov 2026 = A-141 <i>EF</i>	LRR waits on A-134, not on A-133; HP-4
M-7 <i>T-0</i>	15 Nov 2026 = A-143 <i>EF</i>	Baseline (Sec. 1). Option C 10 Nov 2026 is contingent only; not redrawn on Fig. 3.2
M-8 handover	20 Nov 2026 = A-144 <i>EF</i>	Five days after M-7; Table 2.1 closes the 22 Nov collision
Pad occupancy	8 org. / 12 cap / 14 unmit. / 12 mit. (simultaneous unique heads)	Fig. 5.1; Table 5.2; R-13; Sec. 7.4
Surge labour	\$22,400 = $4 \times 7 \text{ d} \times 8 \text{ h} \times \100 inside WBS 1.3.3 (24–30 Oct only)	In the \$1,500,000 base, not a contingency draw
EMV pool	residual $P'\% \times I_{\S} = \$37,920$ (not inherent $P\%$)	Table 4.2; Table 6.1 EMV column
HP-4	Launch Director / RSO / Juru Officer / Abdul / Ziyad; payload sponsor witnesses	Table 7.1; Table 5.1 1.4.1; A1 M-6
Option C	\$21,000 from contingency if R-14 fires	Not in Table 4.3; Board draw under the Sec. 7.1 DoA
Contribution	$25.0\% \times 4 = 100.0\%$ (168 h equalised)	Title page; Table A.1

8.2 Team fallback and contribution

Team-member fallback is a three-tier operating procedure on the work, not a single redistribution bullet. If a member is non-responsive for more than 48 hours: (1) the Lead Planner (Hadi Alkhub) audits the artefacts within 12 hours against the WBS package in Table 5.1; (2) the

Campaign PM (Abdul Wahab Obeid) issues a written addendum reassigning those packages; (3) the Buddycheck Personal Assessment Form is locked against Microsoft Teams and version history so contribution evidence cannot be rewritten after the fact. The procedure reallocates rows in Table 5.1. It does not move $T-0$, BAC , or a hold point. Pad resource conflicts still use the 4 h / 12 h / 24 h SLA in Section 5.3; contingency draws and M-5 / M-7 moves remain Board decisions under Section 7.1.

Contribution is equalised at 42 hours per member so the cover split of 25.0% each remains 100.0%. Table A.1 and Appendix A are that split: Hadi Alkhub on scope and schedule; Ilsa Hashmi on cost, resources and EVM; Ziyad Alahmari on the register, quality and HSE; Abdul Wahab Obeid on the executive summary, integration and final QA. Hours, not seniority, set the percentage.

8.3 Presentation and referencing

The document is 12 point Times on A4, 25 mm margins, 1.5 line spacing, numbered headings to two levels. Figures and tables are numbered within the section, captioned, and referred to in the prose before the float. TikZ draws node diagrams (WBS, governance). Matplotlib reads the YAML register for data plots. Citations are APA 7th author–year through `natbib`. Design choices are cited where they are made: critical path and compression (Project Management Institute, 2021; Kerzner, 2017); hold points (SAE International, 2016); contingency as expected monetary value (Kerzner, 2017; International Organization for Standardization, 2018); labour and plant (Fair Work Ombudsman, 2020; Rawlinsons, 2025); range permissions (Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019); delivery model (Turner, 2014); earned value (Project Management Institute, 2019; Fleming and Koppelman, 2016); this pad’s purpose and TestFlight 1 heritage (Gilmour Space Technologies, 2025; Australian Space Agency, 2025; Dougherty, 2026; Smith, 2025). Word count on the title page excludes references, appendices, and content inside tables and figures.

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A Contribution appendix

Table A.1: Agreed contribution after hours equalisation. Each member 42 h; percentages match the title page and sum to 100.0%.

Team member	Hours	Primary deliverables	%
Abdul Wahab Obeid (s5303839)	42	Executive summary; integration; governance; delivery model; final QA	25.0
Hadi Alkhub (s5331680)	42	Scope; schedule; network logic; Gantt; compression	25.0
Ziyad Alahmari (s5375826)	42	Risk register; quality and HSE	25.0
Ilsa Hashmi (s5310047)	42	Cost; resources; capacity; EVM framework	25.0
Total	168		100.0

Hours are equalised so that the 25.0% shares on the title page remain valid. The three-tier fallback in Section 8.2 (12-hour artefact audit, PM WBS addendum, Buddycheck PAF lock) is the procedure that would reallocate these rows.

We confirm that the contribution percentages above are agreed by all members.

Abdul Wahab Obeid

Signature: _____

Date: _____

Ziyad Alahmari

Signature: _____

Date: _____

Hadi Alkhub

Signature: _____

Date: _____

Ilsa Hashmi

Signature: _____

Date: _____